Wavence 25

Software Installation and NE Upgrade Manual

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Issue 01

Software Installation and NE Upgrade Manual

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Summary of changes

Changes between document issues are cumulative. Therefore, the latest document issue contains all changes made to previous issues.

Changes between release Wavence 24, Issue 01 and Wavence 25, Issue 01

Removed chapters for non-supported HW.

New chapter:

- Downloading and installing a software package (MSS-E/HE/XE and NIM)

Updated chapters:

- Upgrade paths
- Traffic impact during upgrade

Changes between release Wavence 23A SP2, Issue 01 and Wavence 24, Issue 01

Updated chapters:

- Upgrade paths
- Traffic impact during upgrade

Changes between release Wavence 23A, Issue 01 and Wavence 23A SP2, Issue 01

Updated chapters:

- Upgrade paths
- Traffic impact during upgrade

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1. About this document

This guide provides information on software installation, NE upgrades and software rollbacks.

- `SW Migration/Update:` Going from an older release to a newer one. DB and users are migrated too.
  SW Migration/update is done in two steps: Software package (SWP) download and SWP activation.
- `SW Rollback:` Going from a newer release to the previous one existing on this NE. DB and users are back to what was configured in previous state.
- `SW Downgrade:` Going from a newer release to an older that never existed on this NE. DB flash the NE is without configuration. Downgrade is NOT supported.

Depending on your HW equipment/configuration, see the respective chapter:

- CorEvo MSS-4/8: Downloading and installing a software package (CorEvo MSS-4/8)
- MSS-/E/HE/XE and NIM: Downloading and installing a software package (MSS-E/HE/XE and NIM)

For software installation and upgrade on UBT SA, see *Full Outdoor: UBT Standalone (UBT SA) and UBT Nodal Configurations (NIM) User Manual*. For UBT-I SA, see *UBT-I Standalone User Manual*.

> **Note:**
> ETSI modem profiles only support E1 signals and ANSI modem profiles only support DS1/DS3 signals. Mixed configurations (unless explicitly documented) are not supported.

Wavence-E designates ETSI software, and Wavence-A designates ANSI software.

1.1 Official Software Package structure

SW upgrades via WebCT

The following software packages (SWP) are available:

- 1 SWP for CorEvo

- 1 SWP for MSS-E/HE/XE
- 1 SWP for UBT SA (Standalone)

SW upgrades via SNMP

The following software packages (SWP) are available:

- 1 SWP for CorEvo
- 1 SWP for MSS-E/HE/XE
- 1 SWP for UBT SA (Standalone)

For more information on UBT SA, see *Full Outdoor: UBT Standalone (UBT SA) and UBT Nodal Configurations (NIM) User Manual*.

1.2 Used conventions

The generic terms described in Table *Naming conventions* are used when the topic applies to all of the related items listed. If the topic is specific to one or a subset of the items listed, the specific names will be stated in the guide.

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Table 1: Naming conventions

Term	Refers to
MPT-HLS	MPT-HLS standard
MSS	MSS-8
EAS	EASv2 cards
Core	CorEvo cards
CorEvo	CorEvo-1G and CorEvo-10G cards
PDH	E1, DS1, and DS3
SDH	STM-1 and OC-3
UBT	Ultra Broadband Transceiver • UBT-T • UBT-T XP • UBT-S • UBT-m

1.3 Acronyms

Acronyms used in this document.

Table: Acronyms provides the acronyms used in this document.

Table 2: Acronyms

Acronym	Expansion
ATPC	Automatic Transmit Power Control
CT	Craft Terminal
DC	Direct Current
DS1	Digital Signal Level 1
DS3	Digital Signal Level 3
EAC	Ethernet Access Card
EAS	Ethernet Access Switch
EC	Equipment Controller
EPS	Equipment Protection Switch
ETSI	European Telecommunications Standards Institute
EW	Early Warning
FD	Frequency Diversity
FTP	File Transfer Protocol
HSB	Hot Standby
LOS	Loss of Signal
MEF	Metro Ethernet Forum
MPR	Microwave Packet Radio
MPT	Microwave Packet Transport
MSS	Microwave Service Switch
NE	Network Element
NEtO	Network Element Overview
ODU	Outdoor Unit
PDH	Plesiochronous Digital Hierarchy
PFoE	Power Feed over Ethernet

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Acronym	Expansion
QAM	Quadrature Amplitude Modulation
QMA	Quantitative Muscle Assessment
RPS	Receive protection switch
RSL	Receive Signal Level
RTPC	Remote Transmit Power Control
RU	Rack Unit
SD	Space Diversity
SDH	Synchronous Digital Hierarchy
SFTP	Secured File Transfer Protocol
SNMP	Simple Network Management Protocol
SSH	Secure Shell
SW	Software
SWP	Software Package
TAC	Technical Assistance Center
TCA	Threshold Crossing Alert
TCP	Transmission Control Protocol
TPS	Transmit Protection Switch
UBT	Ultrabound Tranceiver
USM	User-based Security Model
XPIC	Cross Polarization Interference Cancellation

1.4 Audience

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Who uses this documentation?

This guide is intended for network installers and system administrators who are responsible for installing, configuring, or maintaining networks. This guide assumes you are familiar with electronic and networking technologies.

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2. SW installation/upgrade information and checks

2.1 Preparation works, precautions and recommendations

Description

The following procedure describes the steps required to upgrade existing Wavence systems with the current release software. New systems are normally delivered with the valid software pre -installed. Reinstallation of the software is not necessary unless the user wants to populate both software banks with the latest software.

For supported upgrade paths, see Upgrade paths.

For part numbers and build information, see the *Release Notice*.

When upgrading, both standard and SNMPv3 (USM) users are migrated to the target release. Password policy is disabled in migration projects whose source release does not support this feature.

The estimated total time to execute the installation procedure is approximately 40 minutes per Wavence shelf. This time does not include any site preparation and pre/post-installation retrievals or setup.

The estimated total time to execute the installation procedure is approximately 60 minutes per Wavence shelf in 1+1 radio protection. This time does not include any site preparation and pre/post-installation retrievals or setup.

Unless explicitly stated, the far end radios from the shelf being upgraded should have the same software release or a release within the direct upgrade path of the near end for guaranteed operation.

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⚠ CAUTION!

Service Disruption

The A16E1DS1 (ASAP) card and ATM services are not supported starting in R6.0.0. If a software upgrade is performed while ATM services are in place over the network, ATM traffic will be lost.

Required materials

The following items are required:

- Wavence system with current equipment as described in the *Release Notice*
- Depending on your configuration and HW, install the relevant SWP inside the SWP CD image (for example: CorEvo, UBT SA)
  The exact version of each software package is described in section *Software Components in Release Notice*.

Precautions and recommendations

Read the entire software installation procedure before beginning.

- This procedure should be performed by a qualified Field Service Engineer trained on the Wavence during an approved maintenance window.
- A copy of the previous release software should be available to assist with any unexpected recovery operation.
- When MSS stacking is in use with LOS switch criteria enabled, disable LOS criteria on both the local NE and the other NE in the stacking configuration before performing the upgrade.

⚠ WARNING!

Traffic and management interruption

This upgrade procedure will cause an extended traffic and management interruption during the activation of the software.

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⚠ CAUTION!

Do not execute a software restart, Core protection switch or create a backup during the software download. The software download will be aborted.

⚠ CAUTION!

Once the current release software has been activated, the current release CT or WebCT must be used to manage the NE.

Note:

Migration of core protected systems should be performed with both Main and Spare cards plugged in.

Note:

Remove all HW in phase-out before activating the software package.

Note:

After upgrading Core-E MSS-8/MSS-4 with LAG L2 Radio, the LAG L2 has to be deprovisioned then reprovisioned.

Note:

In case of MSS-4 10 G with EASv2, upgrade could fail from 7.1/8.0 to W18A or greater release. In that case EASv2 card has to be unplugged and re-plugged.

Note:

Do not upgrade the Network Element Software unless it is known to be required.

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Note:

If an Enhanced Fan W/Alarms (A-FANS) is equipped and configured as an E-FANS, after an upgrade to W18 a Replaceable Unit Type Mismatch alarm condition will be raised. While this alarm is standing, the housekeeping alarm inputs and outputs are not managed.

To avoid this condition, perform one of the following prior to the software upgrade:

Read and understand the following:

1. The existing A-FANS must be reconfigured from E-FANS to A-FANS, OR
2. Install an E-FANS unit which will enable the management of the 8 housekeeping input points.

Note:

Housekeeping alarm configuration parameters will be lost when the E-FANS are reconfigured. It is recommended to record the housekeeping input provisioning before disabling the E-FANS to aid in reconfiguring the housekeeping alarm inputs.

Note:

Forced activations of the software are applied to the MSS but not propagated to the UBT.

Note:

If a flash card in the spare position is older or equal to 8.0 and plugged live on an NE with a higher release, the NE core protection will not be available (e.g. the spare core will not align). The spare core will need to be upgraded to a newer release.

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Note:

In case SW download fails, SW abort is performed from NFM-P and then another SW download or resync of all MIBs is executed, the node remains isolated and SNMP agent does not respond.

To overcome this issue, perform a warm restart of the node from WebCT, being in the same network of NFM-P.

This issue is not present if SW abort is performed from NFM-P during a correct SW download.

Note:

After upgrading from W19A, when only ChA is configured for UBT-T, RDI and PPP fail alarms are raised and traffic hit is present. To overcome this issue, do the following:

`If upgrade is not yet done:`

- Configure the not used channel (ChB) of UBT-T with another frequency: the distance of other channel must be, at least, equal to the CS used by the modem profile. It is not necessary to enable this second channel. After this will be possible to perform the upgrade from W19A.

`If upgrade is already done:`

Two different workarounds are possible:

- To configure the not used second channel disabled (ChB) with another frequency: the distance of other channel must be, at least, equal to the CS used by the modem profile and powered off/on both ends of the link.
- To enable Link Identifier on radio configuration (Interface › Radio Interfaces › Additional Settings) for both channels ChA and ChB of UBT-T using two different IDs and switched off/on both ends.

2.2 Upgrade paths

The following tables show the supported upgrade paths to release 25:

Table 3: Upgrade paths to release 25 (CorEvo)

Source release	Destination release	Possible (Y/N)?	Notes
7.1.0	25	N	Must upgrade to 20A first
19A	25	Y	--
20A	25	Y	
21	25	N	Must upgrade to 21A SP1 first
21A	25	Y	Must upgrade to 21A SP1 first
21A SP1	25	Y	
22	25	N	Must upgrade to 22A first
22A / MPx	25	Y	
23	25	Y	
23A (SP2)/PPx	25	Y	
24 / 24 MPx	25	Y	

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Table 4: Upgrade paths to release 25 (MSS-E/HE/XE)

Source release	Destination release	Possible (Y/N)?	Notes
20A	25	Y	
21	25	N	Must upgrade to 21A SP1 first
21A	25	N	Must upgrade to 21A SP1 first
21A SP1	25	Y	
22	25	Y	
22A	25	Y	Only option valid for MSS-XE
23/ 23MP1	25	Y	
23A (SP2)/ PPx	25	Y	
24 / 24 MPx	25	Y	

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Table 5: Upgrade paths to release 25 (NIM)

Source release	Destination release	Possible (Y/N)?	Notes
21	25	N	Must upgrade to 21A SP1 first
21A	25	N	Must upgrade to 21A SP1 first
21A SP1	25	Y	
22	25	N	Must upgrade to 21A SP1 first
22A/MPx	25	Y	
23/MP1	25	Y	
23A (SP2)/PPx	25	Y	
24/24 MPx	25	Y	

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Table 6: Upgrade paths to release 25 (UBT SA)

Source release	Destination release	Possible (Y/N)?	Notes
19A	25	N	Through WebCT must upgrade first to 20A
20A	25	Y	
21	25	N	Must upgrade to 21A SP1 first
21A	25	N	Must upgrade to 21A SP1 first
21A SP1	25	Y	
22	25	N	Must upgrade to 22A first
22A/MPx	25	Y	
23/ MP1	25	y	
23A (SP2)/ PPx	25	Y	
24/ 24 MPx	25	Y	

Direct migration from releases 19A, 20A, 21A MPx, 22A MPx, 23, 23MP1, 23A, 23A SP2 (PPx) and 24 MPx

For Upgrade limitations, see Limitations during upgrade.

Password policy

Password policy is part of security enhancements.

Password policy is:

- `Disabled by default:`
  - if the system is coming in W19A from a release not supporting it (W18A or older).
  - in green-field deployments
- `Enabled by default` if the system is coming in W19A from release W19 (not possible to deactivate in that release).
- `Aligned with the persistency status of previous release` in case of an upgrade (W19A to W20).

For more information, see *Security* User Manual.

2.3 Traffic impact during upgrade

Traffic impact varies by configuration:

From R7.1.0 to W25

With MPT-HL connected to EASv2: traffic impact due to software migration is around 300 seconds.

From W19A to W25

with UBT-T/S/m connected to EAC or to EASv2: traffic impact due to SW migration: 300 seconds.

with UBT-T/S in 1+1 or 2+2 connected to EAC: traffic impact due to SW migration: 500 seconds.

From W20A to W25

With UBT-T/s/m connected to MSS-E/HE traffic impact: 430 seconds

From W21A SP1 to W25

With UBT-I CA connected to EAC or CorEvo traffic impact: 350 seconds

Upgrade to W25

In case of Radio carrier aggregation: traffic impact is 45 second higher than in other SW migration.

Table 7: Migration from W22 to W25 for UBT-T XP

Configuration	Total Traffic hit
UBT-T XP XPIC intra MSS-8	365 sec
UBT-T XP XPIC intra NIM	280 sec
UBT-T XP 2+2 on EAC	331 sec
CA UBT-T XP on EAC	418 sec

Migration from W23 MP1 to W25 for the CAHD N+N: 600 sec

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2.4 Pre installation checks

In this section, a list of pre installation checks is presented with the corrective action and actions needed in case an issue is not resolved.

> **Note:**
> The following pre-checks should be executed on all nodes of a cluster before executing any action on the nodes.

Table 8: Pre installation checks

Check	Corrective action	Action if NOK
Check Site Name.		
Check Date and Login Time.		
Check Main Core card is in Active Status.	Set Main Core card in Active Status.	Suspend activation.
Check System Settings parameters on CORE protection if in stack with other Wavence: • Check if `StaticLAG`=Enabled • Check if `LOScriteria`=Enabled • Check `ForceCommand`=Disabled • Check `LockCommand`=Disabled	• Enable `StaticLAG`. • Enable `LOScriteria`. • Disable `ForceCommand`. • Disable `LockCommand`.	Suspend activation.
Check CORE protection Mode: COREProtectionMode=Non revertive	Check connectivity structure and Protection Mode.	Suspend activation.
Check for any blocking alarms in NE.	Clear the blocking alarms in NE.	Suspend activation.
Check if the correct SW version is available.	Download the correct SW version.	Suspend activation.
Check all UBT radio are online.	If not, bring all UBT radios online.	Suspend activation.
Check UBT-S/UBT-m LAG	LAGs created using UBT radio links with UBTs is needed to upgrade the nodes in two steps.(nodes from release 18A to 19 Intermediate release then Intermediate to 19A)	Intermediate release is required.

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Check	Corrective action	Action if NOK
Check UBT-T LAG	LAG created using radio links with UBT-T one channel only is needed to upgrade the nodes in two steps. Nodes from release 18A to a 19 Intermediate release then Intermediate to 19A.	Intermediate release is required.
UBT-T both channel A and B are active.	Upgrade directly to final version in one step.	

2.5 Post installation checks

In this section, a list of post installation checks is presented with the corrective action and actions needed in case an issue is not resolved.

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Table 9: Post installation checks

Check	Corrective action	Action if NOK
Node is accessible via the NSP.	NSP should recover the management of the node.	Stop and open a ticket.
Node is upgraded to the target release.	Check final release of the node.	Stop and open a ticket.
Check NE alarms post activation.	If any new alarm comes post activation then update the tracker.	
Unmute all MSS ODUs facing MSS1c.	Check MSS1c is reachable.	Stop and open a ticket **Note:** This action should not be stopped on the rest of ODUs facing MSS1c.
Activate SW on MSS1c.	MSS1c should be reachable after some minutes on the NSP.	Stop and open a ticket.
Disable/Enable all the PM counters for all radios in NE.	Deselect/select options as per guidelines.	Report issues.

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3. SW Package management

The software download operation consists in transferring the SW Package with the programmable element files from an external machine (WebCT or external file server) to the NE.

The Managing System can be the WebCT or a SNMP Manager. Running SWDL over HTTPs and SNMP at the same time is not supported.

In CorEvo and MSS-E/HE/XE configurations, when downloading a new software package (SWP), the ODU verifies that it actually comes from Nokia and was not forged by a malicious hacker. Only a SWP that is signed by Nokia will be used to update the NE. The verification involves:

- Encryption and Hashing algorithm are valid and they are not corrupted either unintentionally (for example memory corruption) or intentionally (for example by an external attacker)
- The SWP comes actually from Nokia (proof of origin).
- The files in the SWP are not corrupted (integrity check).

Beforehand, it is assumed that the ODU has successfully run the self-test of the software routines that implement the algorithms used for the check of:

- Software Integrity (SHA-512)
- Proof of origin / Digital signature check (RSA-4096)

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4. Upgrading and CorEvo protection

Purpose

CorEvo protection is supported starting in R6.1.0 and in all newer releases.

Procedure

1 Log in to WebCT and run the Show Genesys Version command from the Debug Info page. See the *WebCT User Manual* for details.

2 Verify that the version is V1.0.14 or greater. If the version number is not V1.0.14 or greater, the µSD must be replaced before CorEvo protection can be configured.

See the *Release Notice* for µSD card part numbers.

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5. Downloading and installing a software package (CorEvo MSS-4/8)

Purpose

Before you begin

> **Note:**
>
> Do not power off the system during the SW activation procedure.

> **Note:**
>
> In case of SW upgrades and downgrades on MSS equipped with radios powered by the EAC PFoE or QMA, the EAC activation with power off/on of the radios adds 2 minutes.

> **Note:**
>
> The total time to perform the software installation procedures and to download and activate the software on the MSS shelf is around 30 minutes for each ODU. For example, 4 UBTs require a 2-hour upgrade.

> **Note:**
>
> For SW upgrades of NEs with radio LAG links with only UBT-S/m (or UBT-T with one carrier):
>
> These links are subject to radio incompatibility between releases below or equal to W18A and releases equal or higher than W19.
>
> NEs of such clusters must be upgraded within the same night because traffic will be up when radios have the same release.
>
> For any further information, please contact TEC.

Required materials

The following are required:

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- Wavence system with current equipment as described in the *Release Notice*.
- Software load for upgrade, as described in the *Release Notice*.

Precautions and recommendations

Notice:

Failure to observe this safety precautions may result in service disruption.

- Unless explicitly stated, the far end radios from the shelf being upgraded should have the same software release or a release within the direct upgrade path of the near end for guaranteed operation.
- Do not execute a software restart, Core protection switch or create a backup during the software download. The software download is aborted.
- This upgrade procedure will cause an extended traffic and management interruption during the activation of the software.

Note:

- Do not upgrade the Network Element Software unless it is known to be required.
- This procedure should be performed by a qualified technician trained on the Wavence during an approved maintenance window.
- A copy of the previous release software should be available to assist with any unexpected recovery operation.
- When MSS stacking is in use with LOS switch criteria enabled, disable LOS criteria on both the local NE and the other NE in the stacking configuration before performing the upgrade
- Read the entire software installation procedure before beginning.

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Note:

If an Enhanced Fan W/Alarms (A-FANS) is equipped and configured as an E-FANS, a Replaceable Unit Type Mismatch alarm condition will be raised. While this alarm is standing, the housekeeping alarm inputs and outputs are not managed.

To avoid this condition, perform one of the following prior to the software upgrade:

1. The existing A-FANS must be reconfigured from E-FANS to A-FANS, OR
2. Install an E-FANS unit which will enable the management of the 8 housekeeping input points.

Note that housekeeping alarm configuration parameters will be lost when the E-FANS are reconfigured. It is recommended to record the housekeeping input provisioning before disabling the E-FANS to aid in reconfiguring the housekeeping alarm inputs.

Procedure

1 Install the software package on the PC that can access the NE via WebCT.

SW activation of the radios might still be ongoing and there might be alarms due to that (i.e. ICP). Only after some time where SW activation is finished, those alarms that are not gone should be considered as valid. No power off/on at this point before Radios have been active.

2 Perform the following general steps:

- Check both software banks and verify that one of the banks has a version that supports a direct upgrade with an operational status of enabled and a current status of committed.
- Disable 1588 if applicable.
- Check for Alarms and Abnormal Conditions. Resolve all unexpected alarms and conditions prior to the upgrade execution.
- Ensure Main (not Spare) Core card module is active (Protected Core card configurations only).
- Ensure all protection groups are on the Main (non-switched) state.
  If a protection group is enabled on any module, review the status to ensure that the main entity is active. All protection groups should be in a normalized (non-switched) state. If assistance is needed in resolving issues that have caused a protection switch, please contact Nokia. These issues should be resolved before proceeding with this installation.
- Retrieve and save the NE Inventory.
- Perform a system backup (Recommended but Optional).

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- If a spare Core module is equipped, it is recommended that it is unseated at this time. This is not required, but provides a quick recovery option in case of issues during the upgrade. Once the upgrade is successfully completed it can be reinserted to mirror from the main Core. If issues arise during the upgrade unseat the main Core, reinsert the spare Core to recover and contact Nokia for further assistance.

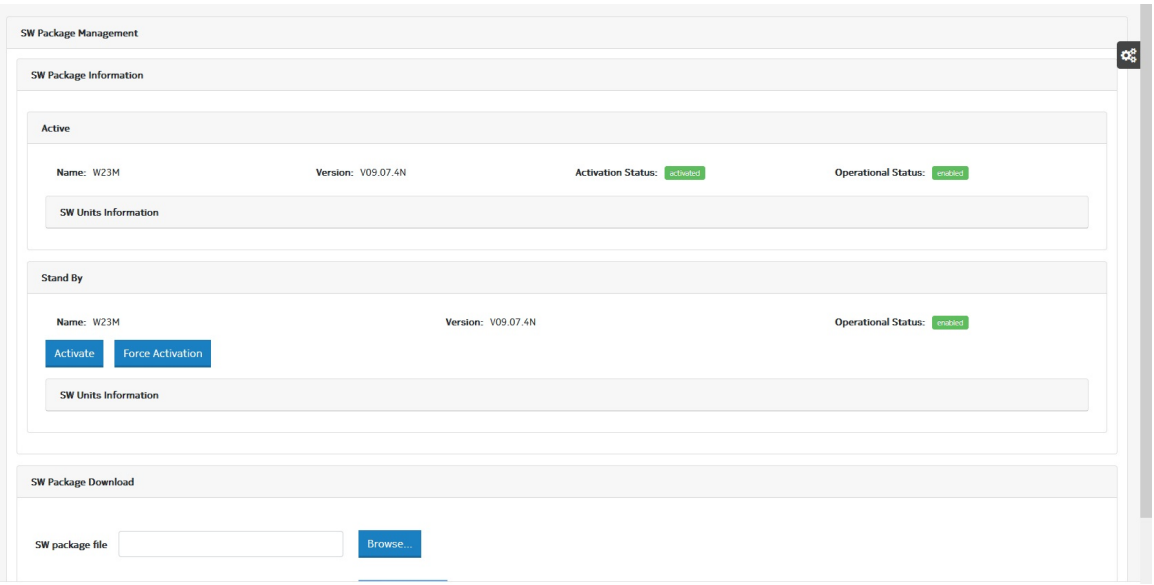
3 From the `Administration & Networking` domain, select `SW Package Management`.

The SW Package Management webpage displays.

On CorEvo (MSS-4/8), Activation Status is also shown.

Activated means that only activation process is completed and you can continue to verify if all radios are online with the new SW and further configuration actions.

Figure 1: SW Package Management (MSS-4/8)



4 To download the software, click the `Browse` button. Navigate to `\PACKAGES\WEBCT` found in the SW installation CD or in the SW package received via Nokia Support Portal.

> **Note:**
>
> For more information on the actual filename of the software package, see *Release Notice*.

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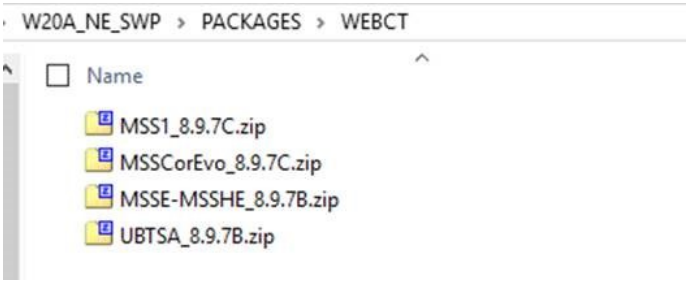

5 Select a valid software package Zip file, and click on open.

Step result

Selected package zip file name is displayed in the `SW package file` field. The `Download` button becomes enabled.

6 Click on the `Download` button to load the software to the shelf.

The zip file is uploaded into the NE and a message is displayed indicating that downloading is in progress. The software download page enters the `loading` state.

There is no indication of percentage progress of the upload.

Once the file has completely been uploaded, in the standby unit area the following message is displayed:

```
A software package download is in progress. You will see the new
package details as soon as the download is completed.
```

> **Note:**
>
> Note: When the Active bank has an earlier release than that of Standby bank, the name of the Standby bank will be R95M.

Once the Firmware download is completed on all HW units, the standby unit operational status changes to the `enabled` status.

> **Note:**
>
> If the new uploaded standby unit's operational status is `disabled`, then HW units or UBT/MPT Firmware download is taking place. DO NOT restart or switch off NE during this time.

Once the Firmware download is completed on all HW units, the standby unit operational status changes to the `enabled` status.

> **Note:**
>
> In case the firmware download is not completed for some IDU units after 30 minutes, the standby unit's operational status changes to `enabled` and the FirmwareDownloadOnGoing alarm is raised. In such case contact the local TAC.

7 Once operational status is changed into `enabled` then user can click on `Activate` or `Force Activate`.

> **WARNING!**
>
> Failure to observe this safety precautions may result in service disruption.
> The software must be activated at this time. Do not restart the system at this time.
> Contact Nokia TAC in the event assistance is required

> **WARNING!**
>
> Failure to observe this safety precautions may result in service disruption.
> If other Wavence shelves are managed through this shelf (via RF link) supervision to those shelves will be lost temporarily during the software activation process. If this operation is being performed via the RF link, it is recommended that personnel be present at the remote site with craft terminal access to the remote NE in the event that supervision to the remote NE is lost.

> **WARNING!**
>
> Failure to observe this safety precautions may result in service disruption.
> Service interruption of up to approximately 10-12 minutes is expected. The interruption may be longer, up to approximately 30 minutes, if SDH is in use.

To activate the software package, click `Activate`.

Difference between Activate and Force Activate:

- Activate: If the same software package version is on the active unit and the standby unit,

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then clicking on the Activate button does not cause a restart of NE and the Active unit remains active.

- Forced Activation: The Forced Activation command is applied independent of the version installed on the two banks. The versions of the software may be the same or different. If the Force Activate button is clicked with the same versions on each unit, the NE restarts and the Standby unit software package becomes activated.

> **Note:**
> Forced activations of the software either from WebCT or SNMP are applied to the MSS but not propagated to the ODU.

Step result

Clicking on Activate or Force Activation produces a warning message.

```
The node will restart as consequence of the forced activation of the SW package.
WARNING: To avoid the risk to damage ODUs, do not switch off any ODU during the
activation.
```

The activation and upgrade process will take approximately 10 to 15 minutes, during which time the shelf will be unavailable for management and traffic may be impacted.

The following message is displayed:

- CorEvo (MSS-4/8):

```
Software package activation began successfully, NE is about to restart.
```

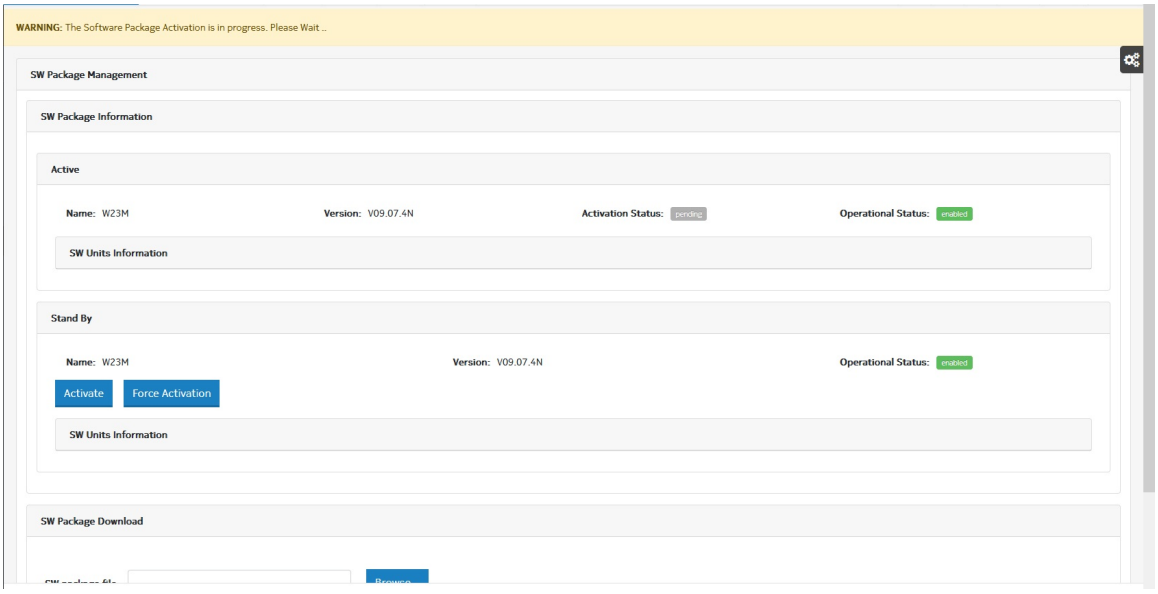
8 Login to the WebCT to verify activation of the Software Package.

CorEvo (MSS-4/8): The warning banner below appears on all pages while the activation procedure is still running.

The Software Package Activation banner refers to any pending firmware download or SW package activation in IDU slots or ODUs.

When the activation procedure is completed, `Activation Status` changes from `pending` to `activated`.

Figure 2: Pending software package activation (MSS-4/8)



9 Check for Alarms and Abnormal Conditions and if OK, load the new Software Package into the Standby software bank by repeating step 3. From the Administration & Networking domain, select SW Package Management. — step 6. Click on the Download button to load the software to the shelf.

10 Perform the following general post-installation procedures:

- Reseat spare Core card if removed.
- Update the License Key (Optional).
- Enable 1588 (Optional).
- Perform a Database Backup of the Upgraded System.

Postrequisites

Note:

To avoid any display browser issues, after upgrading:

1. Clear the browser cache by pressing `Ctrl`+`Shift`+`Delete` together.
2. Refresh the page by pressing `F5` or `Ctrl`+`R`.

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6. Downloading and installing a software package (MSS -E/HE/XE and NIM)

Purpose

To download and install a software package.

Precautions and recommendations

Notice:

Failure to observe this safety precautions may result in service disruption.

- Do not execute a software restart or create a backup during the software download. The software download is aborted.
- This upgrade procedure will cause a temporary traffic and management interruption during the activation of the software.

Note:

- This procedure should be performed by a qualified Field Service Engineer trained on the MSS-E/HE/XE or NIM during an approved maintenance window.
- Read the entire software installation procedure before beginning.

Procedure

1 Save the software package onto the craft terminal PC.

Note:

In case you use Safari, do not use the option to automatically unzip a downloaded zip file. The original zip file must be used.

2 Perform the following general steps:

- Check both software banks and verify that the active SW bank has a version that supports

a direct upgrade with an operational status of enabled.

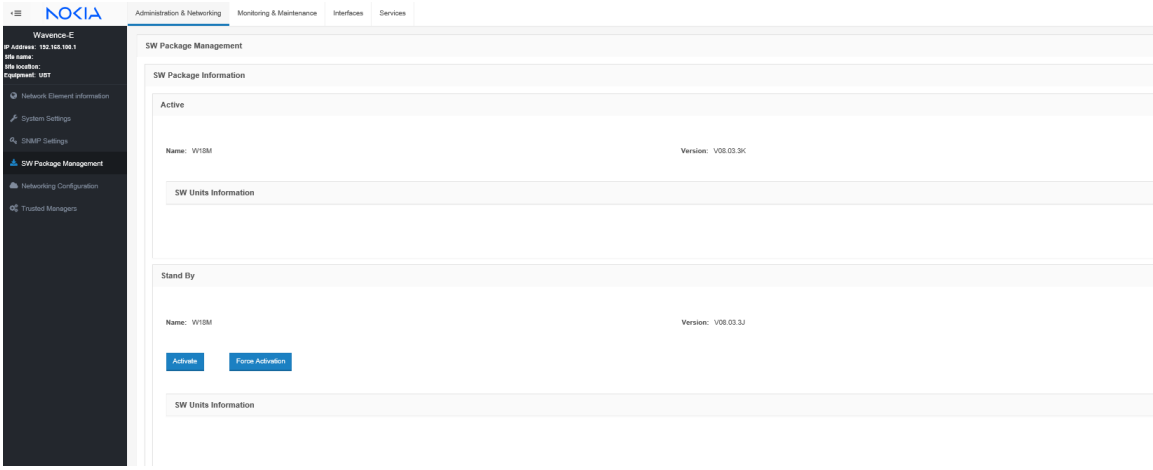
For the migration paths, see *Release Notice* or *9500 MPR/WAVENCE Generic Technical Status (HW/SW Compatibility)* documents.

- Check for Alarms and Abnormal Conditions. Resolve all unexpected alarms and conditions prior to the upgrade execution.
- Perform a system backup (Recommended but Optional).

3 From the `Administration & Networking` domain, select `SW Package Management`.

The SW Package Management webpage displays.

Figure 3: SW Package Management webpage MSS-E/HE/XE and NIM



4 To download the software, download from Nokia Support Portal the ZIP file and store it lockally on your PC. Click the `Browse` button and select the SW package.

> **Note:**
>
> The screenshots below are not release-specific. For more information on the actual filename of the software package, see *Release Notice*.
>
> The actual folder shows all SWP delivered.

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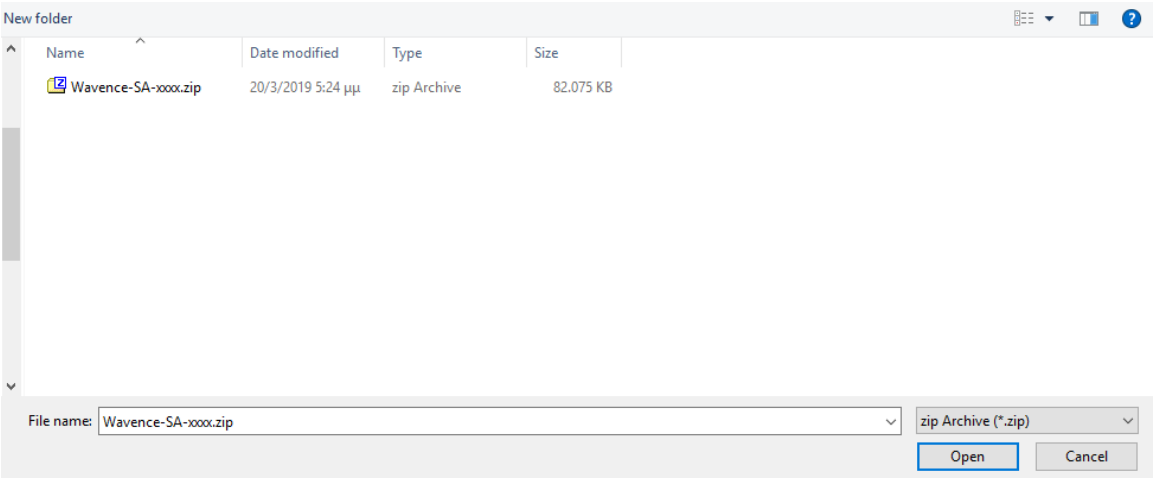
Figure 4: Browsing for the SW package (example)

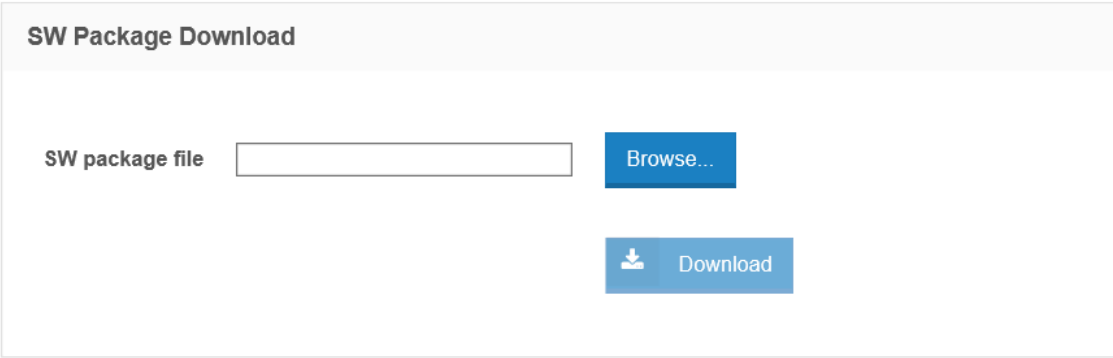


5 Inside the MSS-E/HE/XE or NIM folder, select the .zip file, and click on open.

Step result

Selected package zip file name is displayed in the `SW package file` field. The `Download` button becomes enabled.

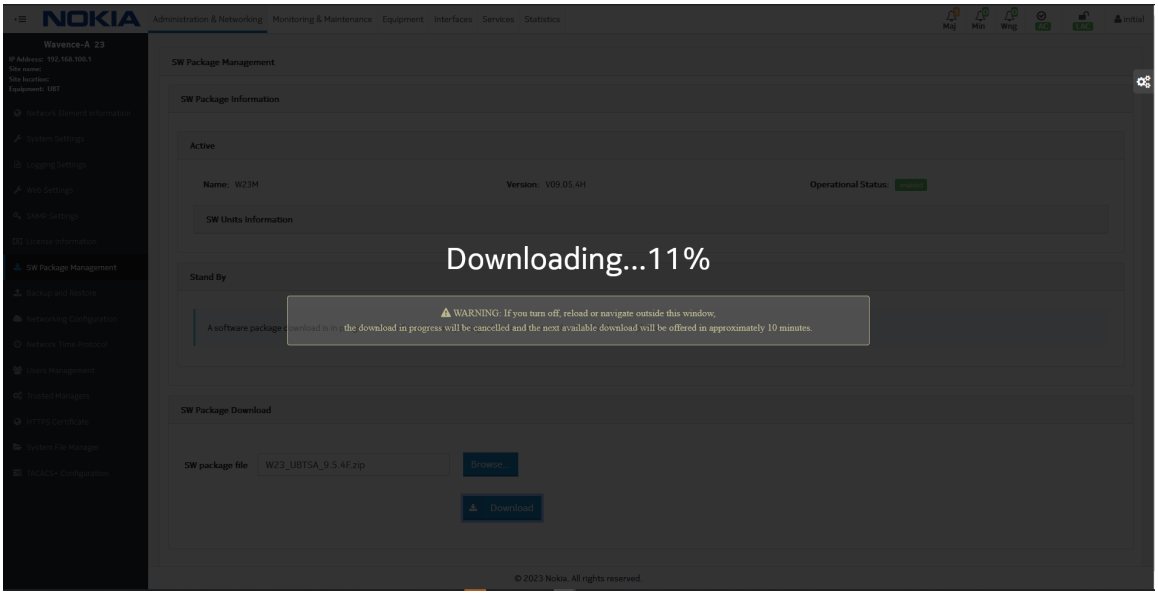
Figure 5: SW package download



6 Click on the `Download` button to start the software download process.

The following page appears warning that if you turn off, reload or navigate outside this window, the download in progress will be cancelled and the next available download will be offered in approximately 10 minutes.

Figure 6: Downloading in progress



The following messages are displayed in case of:

- wrong SWP used: `Failed to start download process. Wrong software package.`
- general failure: `Software package download failed.`
- signature error: `Software package download failed. Signature error.`
- integrity error: `Software package download failed. Integrity error.`
- missing SW unit: `Software package download failed. Software unit missing.`

Once download is completed the following message is displayed:

`Software package download completed successfully.`

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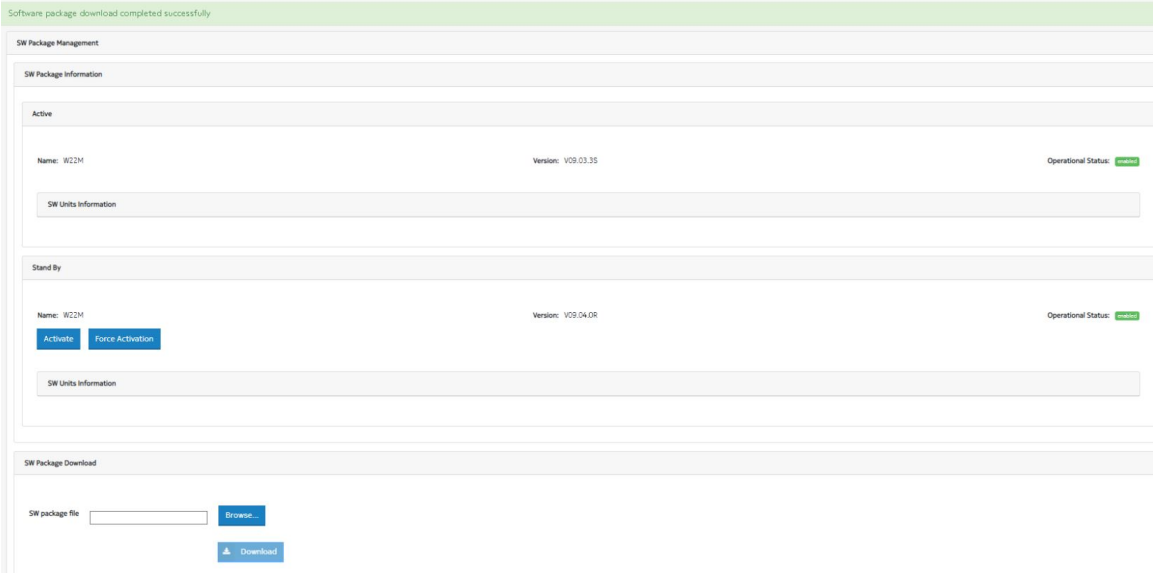
Figure 7: SWP download completion (MSS-E/HE/XE)



The following message is displayed in the background of the SW download progress page or in case the user navigates to SW package management page, while there is SWP download already in progress (either WebCT or SNMP initiated).

`A software package download is in progress. You will see the new package details as soon as the download is completed.`

Once the Firmware download is completed on all HW units, the standby unit operational status changes to the `enabled` status.

7 Once operational status is changed into `enabled` then user can click on `Activate` or `Force Activate`.

In Forced SWP activation, MSS-E/HE/XE or NIM forces the reload of all the units, even of those that have not been changed by the SWP.

WARNING!

Failure to observe this safety precautions may result in service disruption.
Do not restart the system at this time. Contact Nokia TAC in the event assistance is required.
If this operation is being performed via the RF link, it is recommended that personnel be present at the remote site with craft terminal access to the remote NE in the event that supervision to the remote NE is lost.
Service interruption of up to approximately 10-12 minutes is expected. In case of 1+1 Carrier aggregation it is very low.

Step result

Clicking on `Activate` or `Force Activation` produces a warning message.

WARNING: To avoid the risk to damage ODUs or NIM, do not switch any ODU or NIM during the activation.
SWP download takes approximately 15-20 minutes. During this time the FirmwareDownloadongoing alarm is raised ans the SW bank status is disabled.

SWP activation depends on the SW modules that will be reloaded and takes up to a couple of minutes.

The following message is displayed:

Software package activated successfully, Please wait while the NE restarts.

8 Once NE is reachable again, login to the WebCT to verify activation of the Software Package.

9 Perform the following general post-installation procedures:

- Perform a Database Backup of the Upgraded System.

Postrequisites

Note:

To avoid any display browser issues, after upgrading:

1. Clear the browser cache by pressing `Ctrl`+`Shift`+`Delete` together.
2. Refresh the page by pressing `F5` or `Ctrl`+`R`.

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7. Performing a software downgrade

Purpose

To perform a software downgrade.

Rollback is when you are rolling back to the previous SW that was before active but usualy this previous SW is already available in STB bank. So you only need to activate the stb bank and also DB is synced back to the previous release.

> **Note:**
> After a rollback, in case something is wrong (for example not restored traffic), a MSS/Radio power off/on is admitted.

In this section a software downgrade is described. If both banks have been moved in new SW even if you try to download a SW that in the past was there still it is like a downgrade because the DB of that SW was lost. In downgrade case the NE is flashed since the DB is lost.

Before you start

Before downgrading to a version prior to Wavence 18, unseat the EAC board.

Procedure

1 As described in Downloading and installing a software package (CorEvo MSS-4/8):

1.1 Download the SW of a previous release on the NE.

1.2 Check the status and content of the banks.

1.3 Force Activate the SW.

Step result

The previous software package is activated and the NE restarts. The activation and upgrade process takes approximately 10 to 15 minutes. During that time the shelf is unavailable for management and traffic is impacted.

When the NE has finished restarting, access can be regained to the NE.

If the CorEvo runs a SW version supporting WebCT proceed with the steps below. Otherwise, clear the database (DB) and run the alarm checks.

2 Once the NE is up and stable, connect to 10.0.1.2 to open WebCT and verify the activation of the Software Package.

Check the software bank loaded with a previous version and verify that it has:

- `Operational Status`: `Enabled`
- `Current Status`: `Committed`

> **Note:**
> If it is not `Enabled` and `Committed`, do not proceed with this rollback procedure and call Nokia TAC.

3 Verify that all modules are online and alarm free.

If any modules have an `AlarmFirmwareDownloadOngoing` warning active (also blinking status LED), wait for the firmware download to complete before proceeding. `AlarmFirmwareDownloadOngoing` should clear and status LED should be green (or amber if in the spare slot of a protection group). This may take 5 to 15 minutes each. If the module does not clear this alarm, contact Nokia TAC before proceeding.

> **Note:**
> If the system auto scratched, then there may not be any radios downloading firmware.

> **Note:**
> In case the firmware download is not completed for some IDU units after 30 minutes, the standby unit's operational status changes to `enabled` and the FirmwareDownloadOnGoing alarm is raised. In such case contact the local TAC.

4 Make sure the protected core is aligned – if protection is configured.

5 From the WebCT `Monitoring & Maintenance` page, go to `Debug Info` and run the

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command `Clear DB and Restart NE`.

6 Once the NE is up and stable, connect to 10.0.1.2 to open WebCT.

7 Make sure the protected core is aligned – if protection is configured.

8 Power cycle the entire shelf.

9 Check for Alarms and Abnormal Conditions. No downloading alarm present.

Postrequisites

Note:

To avoid any display browser issues, after roll-backing to a previous Wavence release:

1. Clear the browser cache by pressing `Ctrl`+`Shift`+`Delete` together.
2. Refresh the page by pressing `F5` or `Ctrl`+`R`.

8. Configuring PC network card to connect to NE for MSS-4/8

Purpose

This example uses a Microsoft Windows 7 system.

Procedure

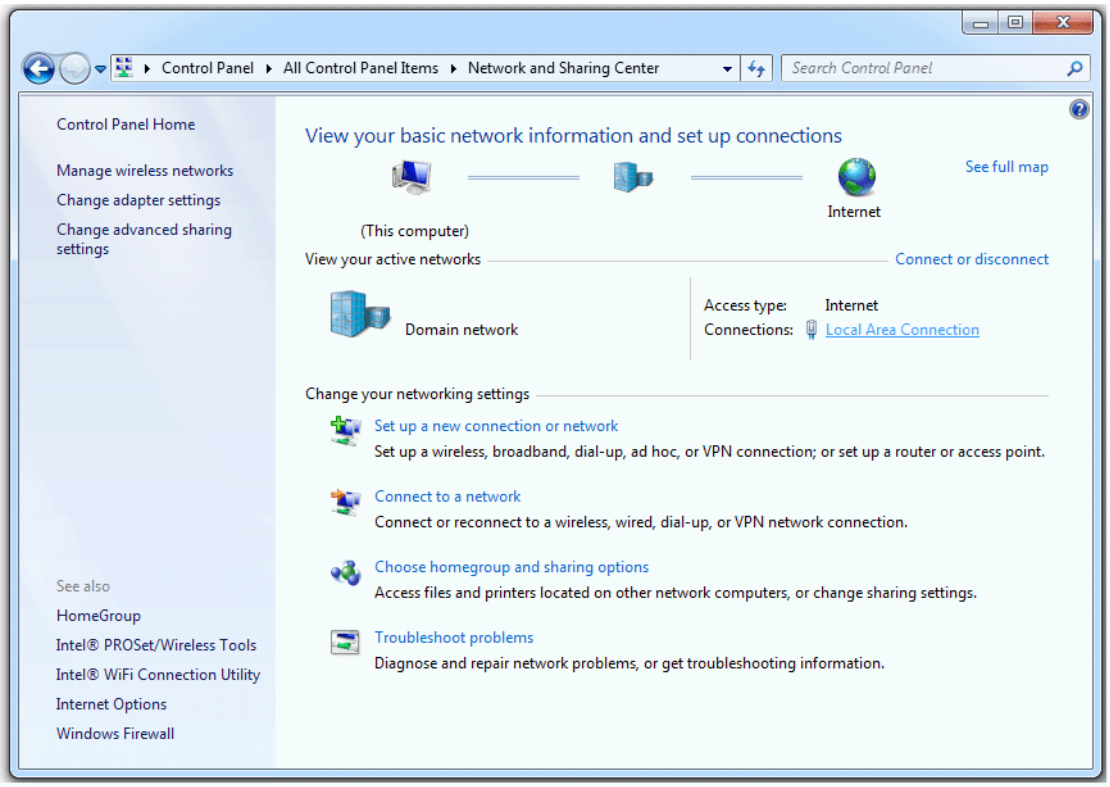
1 Connect a CAT 5/5E cable from the PC network card to NMS connector on the Slot 1 Core card.

2 Click on the START menu on the Windows desktop and open the Control Panel.

3 Open the Network and Sharing Center.

Step result

Figure 8: Network and Sharing Center



4 Click on Local Area Connection (right/center screen).

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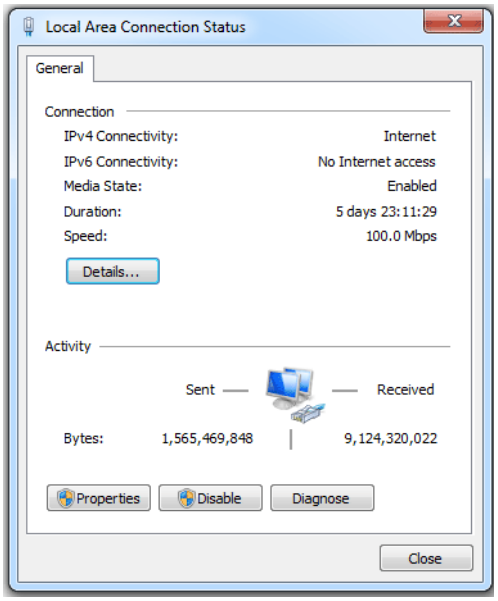
Step result

The Local Area Connection Status window opens.

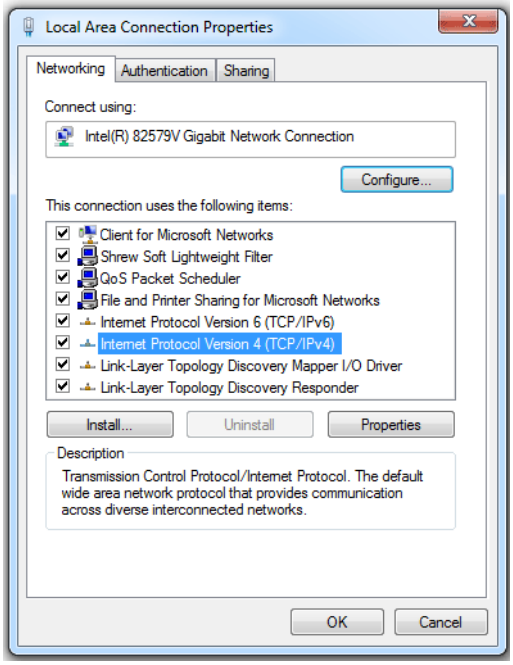
Figure 9: Local Area Connection Status



5 Click on Properties to display the Networking tab.

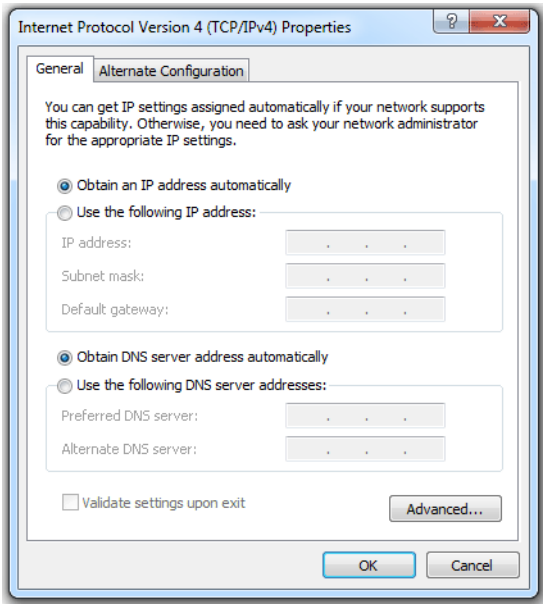
6 Scroll down the list and highlight the Internet Protocol Version 4 (TCP/IP) line. Then click on the Properties button.

Figure 10: Local Area Connection Properties



7 By default the DHCP server on the Wavence is enabled . Set the PC to obtain an IP address automatically.

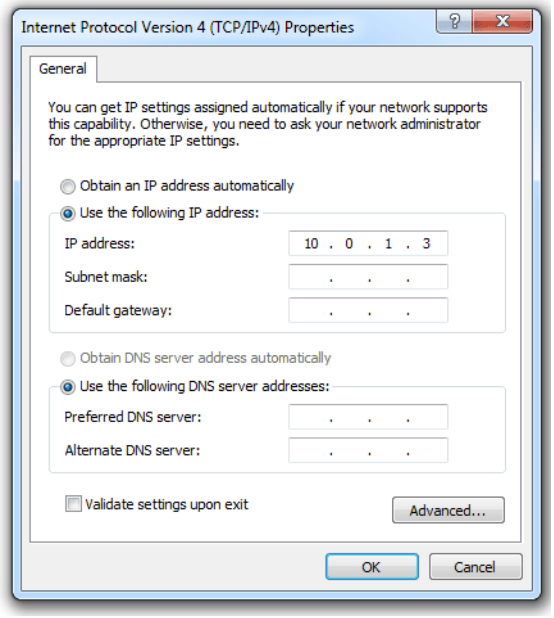
Figure 11: IPv4 Properties with DHCP enabled



!

If for any reason the DHCP server on the Wavence has been disabled, enter the IP address for the PC network card

Figure 12: IPv4 Properties with IP address entered



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The 10.0.1.3 IP address example shown is derived from the default NE IP address for the MSS-4/8 (10.0.1.2) plus 1.

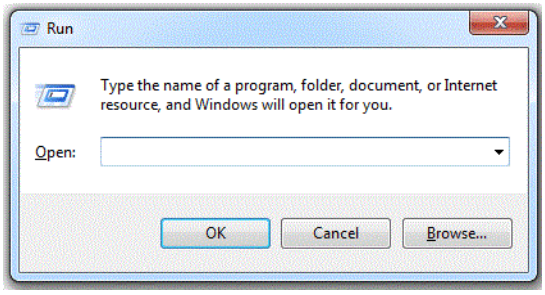
If there is an IP address conflict within your network, increase the last number by two.

> **Note:**
>
> It is suggested to keep only one network connection enabled on a PC.

Click OK.

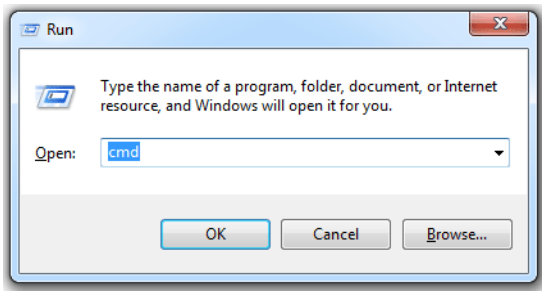
8 To check the connectivity between the PC and the NE, open a DOS window or a Command Prompt.

To get to a DOS prompt in Windows, click on the START menu on the Windows desktop and type RUN in the window on the bottom to open a RUN dialog box.



9 Type cmd and click OK to open a DOS window.

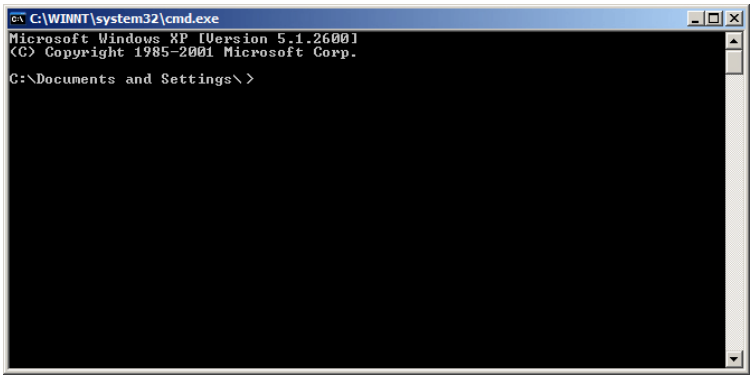
Figure 13: Run window



Step result

The DOS window is displayed.

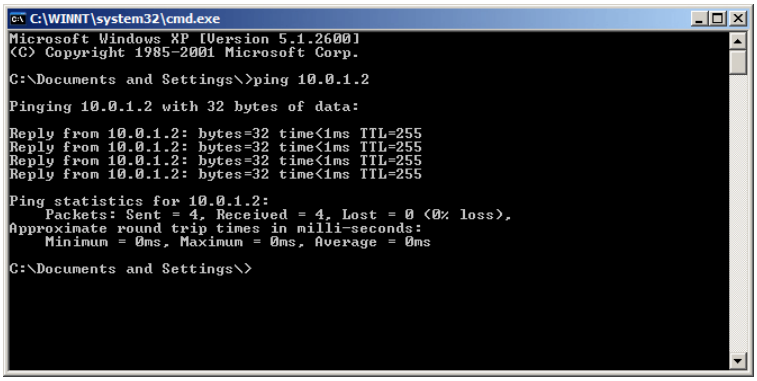
Figure 14: DOS window



10 In the DOS window, click the cursor after the > and type ping 10.0.1.2 to verify a connection between the PC and the NE. The Ping statistics for the IP address 10.0.1.2 should display 4 packets sent and 4 packets received.

> **Note:**
>
> The 10.0.1.2 IP address is the default NE IP address.

Figure 15: ping 10.0.1.2



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9. Limitations during upgrade

This chapter presents some known open issues that occur during upgrade from all releases, along with their workarounds.

Table 10: Upgrade limitations

Limitation	Description	Workaround
After upgrading to W25, in MSS-4/8 with SNMPv3 mode.	Previously existing Trusted Managers' configuration is lost.	Reconfigure any previously configured Trusted Managers.
On a network using UBT-m.	1588 TC Fail Alarm is raised and remains open during migration.	Wait until both local and remote NEs are migrated to the same release.
PDH traffic hit	After migration with existing PDH cross-connections or newly created ones, power off/on or PDH board plug out/in in the new release, PDH traffic hit may occur.	Delete and recreate those cross-connections.
SW rollback should be allowed in the case of license key provisioning mismatch.	If there is a standing provisioning mismatch alarm due to license key mismatch, software rollback is not allowed.	Upgrade the Key before rolling back the software.
MPT at reference modulation after roll-back to release without 1588 capability	If a radio link is upgraded to 7.1 and the user rolls-back to an older release without 1588 HW capability (6.1 or older), the modulation format will still be set to Reference Modulation due to the 1588 calibration process that is still pending.	Pay attention to the reference modulations and adjust accordingly if needed.
Updated modem profile for the ETSI market for the UBT in 112 MHz channel spacing.	This update replaces the existing modem profile and allows 4096 QAM across the entire supported temperature range. The new profile breaks radio compatibility with the current 112 MHz profile in Wavence 19 and earlier. During the upgrade process that adaptive modulation cannot be used with 112 MHz channel size	Change the modulation to a fixed modulation (4QAM, 256QAM, 512QAM, 1024QAM, or 2048QAM) during the upgrade. Once both ends of the link have been updated to Wavence 19A or 20, adaptive modulation can be reconfigured.

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Limitation	Description	Workaround
18Dir: MPT StbyVersionMismatchAlm remain on after SWP Download	In some rare cases the StbyVersionMismatch alarm remain active on some MPT -HL after a software download.	Restart the NE.
UBT-m modem profile introduction (512 QAM) in 250 MHz and 500 MHz	Air compatibility between release 18A and 19 is not supported.	In case of SW upgrade, first upgrade the remote site.
upgrade from 18A UR to 19, UBT connected on Electrical port of EAC or CorEvo	Sporadically UBT is not up.	Power off/on.
After upgrade from 18A	Peak and Average PM is not automatically enabled	Check and enable the required PM manually.

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